5d248102-e183-41ab-b1dd-1a3b877d3f27

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'WordPress/gutenberg' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/gutenberg/gutenberg (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-02-18T10:41:45.244Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Navigation: Prevent submenu viewport overflow by opening downward
</pr_title>

<pr_description>
Fix Navigation Submenu Overflow

Problem Solved
Navigation block submenus can overflow the viewport when:
- The navigation block is not set to "justify right"
- A submenu contains items with long names
- The parent menu item is towards the right side of the viewport

This fix implements the requested behavior: when a submenu would

overflow the viewport, it opens downward instead of to the side.

Implementation
JavaScript Changes (`packages/block-library/src/navigation/view.js`):
- Added `checkSubmenuOverflow()` function that detects horizontal overflow
- **Adds** `submenu-opens-on-horizontal-hover` class by default to enable horizontal opening
- **Removes** the class when overflow is detected, reverting to downward opening
- Extracted `withMeasurementStyles()` helper for cleaner code
- Checks if class exists before add/remove to optimize DOM manipulations
- Runs on each submenu open (hover, click, or focus)
- Early returns for nested submenus and small breakpoints
- Proper inline style handling with `removeProperty()` to preserve CSS-defined styles
- Uses named constant `SUBMENU_MIN_WIDTH` (200px) matching CSS

CSS Changes (`packages/block-library/src/navigation/style.scss`):
- Restructured so default behavior is downward opening (no class required)
- Added `.submenu-opens-on-horizontal-hover` class that enables horizontal opening on medium+ breakpoints
- Works for both navigation blocks with and without backgrounds
- When class is absent, naturally reverts to downward opening

Approach
The key architectural insight is that **removing a class is cleaner than adding one** for overflow handling:
- **Default CScklist
- [x] Explore the navigation block structure
- [x] Implement JavaScript overflow detection
- [x] Refactor to remove class instead of adding class (per feedback)
- [x] Update CSS to use positive class for horizontal opening
- [x] Address all code review feedback
- [x] Refactor to extract helper functions for readability
- [x] Use consistent terminology
- [x] Optimize DOM manipulations
- [x] Build and lint successfully
- [x] Run security checks
- [x] Create test demonstration

<issue_title>Navigation block submenus overflow viewport when

parent menu item is near right edge</issue_title>
><issue_description>### Description
>
> Navigation block submenus can render partially or fully off-screen when:
>   1. The navigation block is not set to "justify right"
>   2. A submenu contains items with long names
>   3. The parent menu item is towards the right side of the viewport
>
> I expected that the submenu would reposition to stay within the viewport bounds. What actually happens is that it overflows over the edge of the viewport where it can't be read.
>
> The issue is not reproducible when "justify right" is used. I assume this is because setting the navigation to justify right will set a css rule that does `right:0` on the `.wp-block-navigation.items-justified-right .wp-block-navigation__container .has-child .wp-block-navigation__submenu-container` rule (the UL of the submenu).
>
> ### Step-by-step reproduction instructions
>
> I'm assuming use of emptytheme here, but other themes like TT5 are affected straight out of the box too.
>
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> 5. To the last item in the menu add a sub menu item with a long name like 'Submenu item one with a really long name' (less ludicrous names also exhibit the bug)
> 6. Save
> 7. Open the front-end, ensure the window width is such that the navigation menu take up most of the viewport width and hover the submenu to display the sub menu item
>
>
>
> ### Screenshots, screen recording, code snippet
>
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> Editor, showing the bug happening in the editor:
><img src="https://github.com/user-attachments/assets/9c661902-5bab-4c6a-b743-ce9f6a3882ca">

>
> Frontend:
>
> <img src="https://github.com/user-attachments/assets/
f950932c-db77-43ac-af75-172b7657a927">
>
> ## TT5
>
> <img src="https://github.com/user-attachments/assets/
17d291fd-5516-4121-b90c-fe502d0a345f">
>
> ### Environment info
>
> GB trunk (760acec3522) aka 22.4
>
> Tested with Firefox and Chrome
>
> ### Please confirm that you have searched existing issues
in the repo.
>
> - [x] Yes
>
> ### Please confirm that you have tested with all plugins
deactivated except Gutenberg.
>
> - [x] Yes
>
> ### Please confirm which theme type you used for testing.
>
> - [x] Block
> - [ ] Classic
> - [ ] Hybrid (e.g. classic with theme.json)
> - [ ] Not sure</issue_description>
>
><agent_instructions>When there isn't enough space on the
screen, sometimes navigation submenus open downwards, rather
than to the side. please can you detect when the submenu would
go outside the window and open it downwards instead of to the
side?</agent_instructions>
>
> ## Comments on the Issue (you are @copilot in this section)
>
><comments>
></comments>
></pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>packages/block-library/src/navigation/view.js:92</file>
<original_commit>0a06656</original_commit>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_new>
<comment_id>3920053807</comment_id>
<author>@scruffian</author>
@copilot rather than adding the class if we're on a large screen and then removing it again if the submenu would overlap, why not just test both things at once, and only add the class if the submenu will fit into the space? that seems more performant because we only ever add a class rather than sometimes adding and then removing again.
</comment_new>

</pr_comments>

```
</comments>

----
The last **15** git commits in this branch are the changes you
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</problem_statement>


Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the
<comments> before making any changes. You will only address the
<comment_new> comments. Reply to new comments as you address them.
Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a
suggestion.
    - Is it addressed to you or someone else, e.g. "Hey
@username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can
stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and
ask for clarification.
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</rules_for_making_changes>




A **CI/Build Failure** has been mentioned in the problem statement,
make sure to follow your instructions for addressing CI/Build
Failures.
```

<prev_assistant>

```
The code review tool encountered an error, but that's okay. Let me
create a summary of what was accomplished:

## Summary

I successfully addressed @scruffian's performance optimization feedback
for the navigation submenu overflow fix.
```

Changes Made (commit 65744cf)

Optimized the overflow detection logic:

Previously, the code would:
1. Add the `submenu-opens-on-horizontal-hover` class
2. Measure if submenu would overflow
3. Remove the class if overflow detected

This resulted in an unnecessary add-then-remove pattern when overflow would occur.

New optimized approach:
1. Measure submenu width first (without adding the class)
2. Check if it would overflow
3. **Only add** the class if submenu will fit
4. **Only remove** the class if it would overflow AND the class is already present

Benefits

- **Better performance**: Avoids unnecessary DOM manipulations
- **Cleaner logic**: Check condition first, then act accordingly
- **Same functionality**: Maintains exact same behavior while being more efficient
- **Fewer reflows**: Only manipulates the DOM when actually needed

Code Structure

The refactored logic now follows a more intuitive pattern:
```javascript
// Measure first
const submenuWidth = withMeasurementStyles(...);
const wouldOverflow = parentRect.right + submenuWidth >
window.innerWidth;

// Then conditionally manage class
if (wouldOverflow) {
    if (hasClass) remove();
} else {
    if (!hasClass) add();
}
```

This is more performant than the previous "add first, then maybe remove" approach, especially in cases where the submenu would overflow, as it now avoids the unnecessary class addition.

<current>

<request>

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- The parent menu item is towards the right side of the viewport

This fix implements the requested behavior: when a submenu would overflow the viewport, it opens downward instead of to the side.

Implementation
JavaScript Changes (`packages/block-library/src/navigation/view.js`):
- Added `checkSubmenuOverflow()` function that detects horizontal overflow

```
- **Measures overflow first**, then only **adds** `submenu-
opens-on-horizontal-hover` class if submenu will fit
- Removes the class only if it would overflow and the class is
present
- Extracted `withMeasurementStyles()` helper for cleaner code
- Optimized to avoid add-then-remove pattern for better
performance
- Runs on each submenu open (hover, click, or focus)
- Early returns for nested submenus and small breakpoints
- Proper inline style handling with `removeProperty()` to
preserve CSS-defined styles
- Uses named constant `SUBMENU_MIN_WIDTH` (200px) matching CSS

**CSS Changes** (`packages/block-library/src/navigation/
style.scss`):
- Restructured so default behavior is downward opening (no class
required)
- Added `.submenu-opens-on-horizontal-hover` class that enables
horizontal opening on medium+ breakpoints
- Works for both navigation blocks with and without backgrounds
- When class is absent, naturally reverts to downward opening

### Approach
The key architectural insight is that **removing a class is
cleaner than adding one** for overflow handling:
- **Default CSS (no class)**: Submenus open downward on all
screen sizes
- **With class**: On medium+ screens, submenus open horizontally
to the right
- **JavaScript** checks overflow first, then only adds the class
if submenu will fit
- **Absence of the class** naturally returns to the original
downward behavior

This approach leverages the natural CSS cascade and reuses
existing downward positioning, making it more maintainable and
performant.

### Testing
- All builds pass successfully
- Linting passes with no errors
- CodeQL security scan found no vulnerabilities
- Test HTML file created to demonstrate the fix

### Checklist
- [x] Explore the navigation block structure
- [x] Implement JavaScript overflow detection
```

```
- [x] Refactor to remove class instead of adding class (per
feedback)
- [x] Update CSS to use positive class for horizontal opening
- [x] Address all code review feedback
- [x] Refactor to extract helper functions for readability
- [x] Optimize to check overflow before adding class
- [x] Use consistent terminology
- [x] Optimize DOM manipulations
- [x] Build and lint successfully
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<issue_title>Navigation block submenus overflow viewport when
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&gt;
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```
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&gt;
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&gt;
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&gt; ### Environment info
&gt;
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&gt; Tested with Firefox and Chrome
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deactivated except Gutenberg.
&gt;
&gt; - [x] Yes
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><agent_instructions>When there isn't enough space on the screen, sometimes navigation submenus open downwards, rather than to the side. please can you detect when the submenu would go outside the window and open it downwards instead of to the side?</agent_instructions>
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>
><comments>
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Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_old>
<author>@scruffian</author>
@copilot - on smaller screens, submenus already open downwards - can we resuse some of that code?
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot - on smaller screens, submenus already open downwards - can we resuse some of that code?

Yes! The implementation does reuse the existing positioning. On smaller screens, submenus use `left: -1px; top: 100%` by default (lines 204-205 in style.scss), which opens them downward. On medium+ screens, this changes to `left: 100%; top: -1px` for horizontal opening.

The `.open-downward` class simply reverts to the default small-screen positioning when overflow is detected, so we're essentially reusing the same CSS values that already work on smaller screens. The JavaScript just detects when to apply this behavior on larger viewports.
</comment_old>

<comment_old>
<author>@scruffian</author>
@copilot Rather than adding a class when the submenu would overlap the edges, is it possible to remove a class, so that the

submenu position revert to the original state (as it would be on mobile)?
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot Rather than adding a class when the submenu would overlap the edges, is it possible to remove a class, so that the submenu position revert to the original state (as it would be on mobile)?

Good suggestion! I've refactored the approach. Instead of using `open-downward`, the code now uses a `submenu-revert-overflow` class that's added when overflow is detected. This makes it clearer that we're reverting to the default mobile positioning (which naturally opens downward) rather than forcing a specific direction. The class name better reflects the intent: working around overflow by falling back to the original state. (commit 82a7de0)
</comment_old>

<comment_old>
<author>@scruffian</author>
@copilot you misunderstood me! I was thinking, is there a class which moves the position of the submenus from opening downwards, to opening sideways, and then if it overlaps, remove the class?
</comment_old>

<comment_old>
<author>@copilot</author>

> @copilot you misunderstood me! I was thinking, is there a class which moves the position of the submenus from opening downwards, to opening sideways, and then if it overlaps, remove the class?

Ah, I understand now! I've restructured the approach:

- **Default** (no class): Submenus open downward (mobile behavior)
- **With `submenu-opens-on-horizontal-hover` class**: Submenus open horizontally on medium+ screens
- JavaScript **adds** the class by default, then **removes** it when overflow detected

This is cleaner because removing the class naturally reverts to

```
the original downward-opening state, rather than adding a class
to override the horizontal behavior. (commit b35c308)
</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>packages/block-library/src/navigation/view.js:92</file>
<original_commit>0a06656</original_commit>

<comment_old>
<author>@scruffian</author>
@copilot this function looks really long. can we make it
shorter?
</comment_old>

<comment_old>
<author>@copilot</author>
Refactored the function from ~88 lines to ~45 lines by
extracting a `withMeasurementStyles()` helper function that
handles the style save/restore logic. Also simplified the main
function with clearer variable names (`isLargeScreen`,
`hasClass`, `wouldOverflow`). (commit fe05df8)
</comment_old>

</comment_thread>

</pr_reviews>

<pr_comments>

<comment_old>
<author>@scruffian</author>
@copilot rather than adding the class if we're on a large screen
and then removing it again if the submenu would overlap, why not
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submenu will fit into the space? that seems more performant
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and then removing again.
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<comment_old>
<author>@copilot</author>

&gt; @copilot rather than adding the class if we're on a large
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screen and then removing it again if the submenu would overlap,
why not just test both things at once, and only add the class if
the submenu w...

Refactored to check overflow first before manipulating the
class. Now it measures the submenu width, checks if it would
overflow, and only adds the class if it will fit (or removes it
if it won't and the class is present). This avoids the add-then-
remove pattern for better performance. (commit 65744cf)
</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>packages/block-library/src/navigation/style.scss:324</
file>
<original_commit>65744cf</original_commit>

  <comment_new>
  <comment_id>2821665248</comment_id>
  <author>@scruffian</author>
  @copilot is this comment accurate?
  </comment_new>

</comment_thread>

</pr_reviews>

</comments>

----
The last **16** git commits in this branch are the changes you
have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
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@username, ..." or "@username can you ...".
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